STAFF REVIEW · H446 262 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The data at memory address 30.

2. A CPU register holding an arithmetic/intermediate value.

3. A readable abbreviation of an operation code.

4. low0 high8 mid4→13, low5 high8 mid6→19; found index 6.

I Do — source worked steps

1. Predict — Store the first input 3 into NUM so the second input will not overwrite it.

2. Explain — Read the second input 5, add NUM to get 8 and output the result.

We Do — guided CHECK

1. Predict
Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.
Explain
LDA A restores the first input 9 and SUB B produces 5; OUT prints 5.

You Do — source mark schemes

1. [4] INP takes input to ACC [1]; OUT outputs ACC [1]; LDA loads ACC from mailbox [1]; STA stores ACC to mailbox [1].

2. [6] First INP makes ACC6 [1]; STA X stores X6 [1]; second INP makes ACC8 [1]; ADD X makes ACC14 [2]; OUT prints 14 [1].

3. [6] INP; STA A [1]; INP; STA B [1]; LDA A [1]; SUB B [1]; OUT; HLT and A DAT 0; B DAT 0 [1]; output 7 [1]. Assume inputs and the non-negative result are representable in LMC's three-digit mailboxes.

Exit Ticket — source answer

1. False. STA X copies ACC into memory X; LDA X loads the content of X into ACC.